UNIVERSITY OF ZULULAND
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# Decision-Grade Multi-Objective Benchmarking and Emulation of Controller Placement in Software-Defined Networks

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*"Driving Sustainable Futures Through Interdisciplinary Research & Digital Innovation"*

## 1 Background & Motivation

SDN ARCHITECTURE

Software-Defined Networking (SDN) decouples control logic from physical forwarding hardware. In multi-site WANs, optimizing the **Controller Placement Problem (CPP)** is an *NP-hard* facility location challenge balancing propagation latency, fault tolerance, and computational cost.

**The Methodological Gap:** Deep Reinforcement Learning (DRL) is frequently advocated for CPP. However, existing studies evaluate models on single-metric latency under isolated conditions, ignoring computational energy costs, training runtimes, and real-world OpenFlow validation.

## 2 Research Questions

OBJECTIVES

**RQ1 (Performance & Latency):** How do DRL algorithms (PyTorch DQN) compare against classical heuristics (*K*-Means, Greedy *K*-Center) in minimizing propagation latency $L(P)$ and maximizing $N{-}1$ reliability?

**RQ2 (Complexity Tax $\omega$):** What computational overhead (s/episode) is incurred by AI models across varying topology scales ($N$ = 20, 50)?

**RQ3 (Hardware-in-the-Loop Emulation):** Do algorithmic rankings translate to real OpenFlow control Round Trip Times (RTT) in Mininet?

## 3 5 Placement Paradigms & Pipeline

METHODOLOGY

**1. Random Baseline** `O(1)`
Stochastic uniform node selection baseline.

**2. Greedy K-Center** `O(k·N)`
Gonzalez 2-approx minimax radius heuristic.

**3. K-Means Medoids** `O(I·k·N)`
Graph geodesic shortest-path partitioning.

**4. Genetic Algorithm** `O(G·P)`
Combinatorial evolutionary crossover search.

**5. PyTorch Deep Q-Network (DQN)** `DRL Model`
Adaptive neural placement (64×64 MLP), $\gamma$ = 0.99, $\alpha$ = 0.001, replay buffer $10^4$, $\epsilon$-greedy decay.

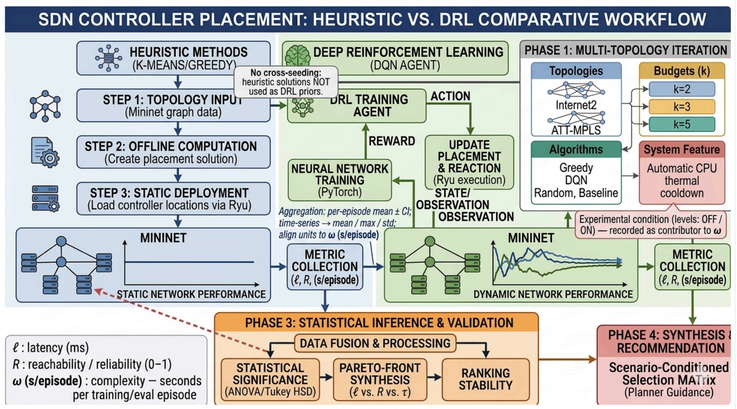



*Fig. 1. End-to-end multi-objective benchmark and Mininet OpenFlow emulation pipeline.*

| 2.852 ms | < 5 ms | 100 % | 20 Ep. |
|---|---|---|---|
| MININET RTT (K-MEANS) | HEURISTIC COMPUTE | N−1 LINK RELIABILITY | DQN CONVERGENCE |

## 4 3-Axis Multi-Objective Pareto Space

TRADE-OFF ANALYSIS

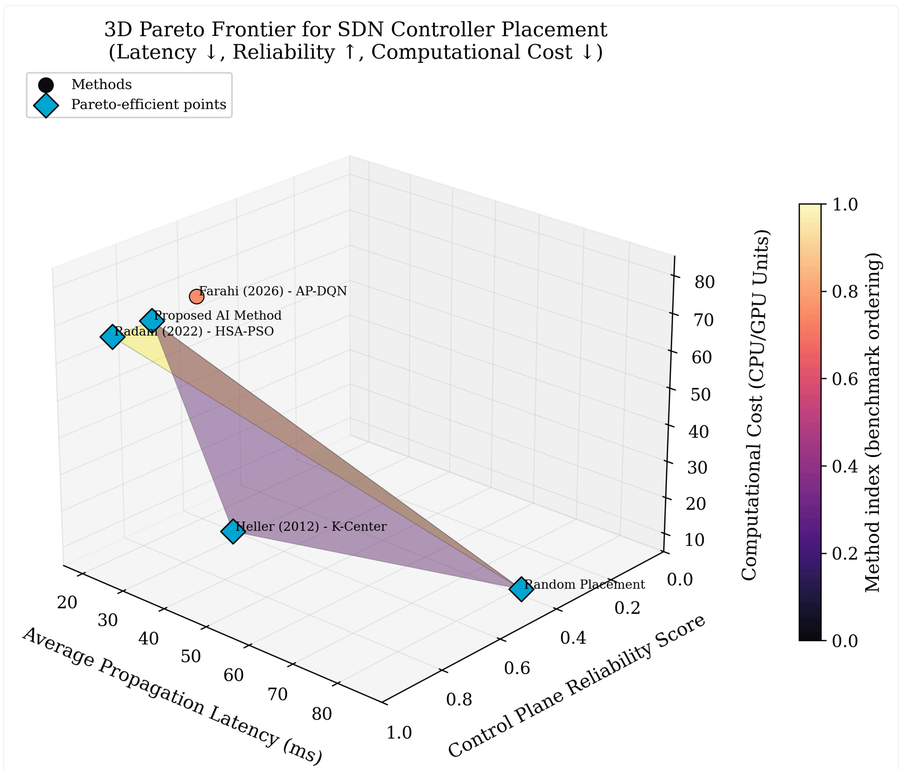



*Fig. 2. Joint 3-Axis Pareto space: Latency $L(P)$ (ms) vs $N{-}1$ Reliability $\text{Reach}_{\text{avg}}$ (%) vs Compute Time $\omega$ (s).*

The non-dominated Pareto frontier proves that **K-Means Clustering** and **Greedy K-Center** achieve the optimal trade-off boundary—delivering minimal latency with near-zero compute overhead ($\omega$ < 0.005 s).

## 5 DQN Training & Policy Dynamics

1,000 EPISODES

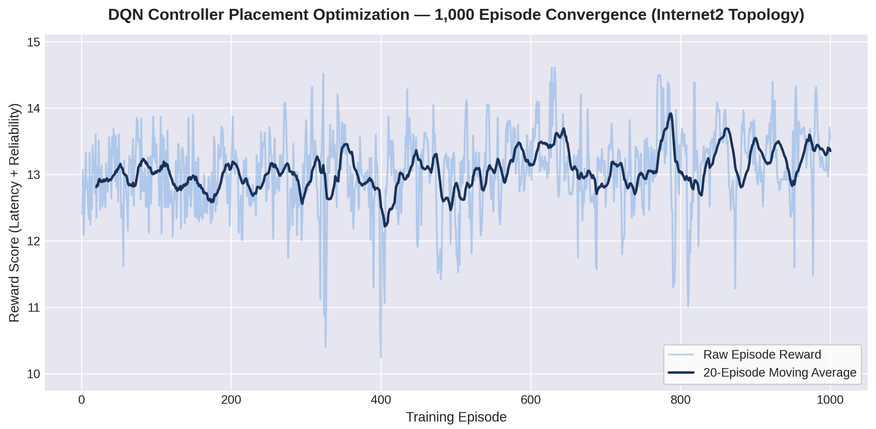



*Fig. 3. 1,000-episode DQN reward trajectory on Internet2 ($N$ = 34, $k$ = 3), converging within 20 episodes.*

DQN converged rapidly to a mean reward of **13.199** (total runtime: 21.26 s, $\omega$ = 0.0213 s/episode). However, it incurs a **4,000× computational tax** over heuristic clustering without achieving lower propagation latency.

## 6 Mininet OpenFlow Emulation

EMULATED VALIDATION

Hardware-in-the-loop control RTT was measured in real-time across virtual Open vSwitch networks on Internet2 WAN ($N$ = 34, $k$ = 3):

| ALGORITHM PARADIGM | CONTROL RTT | COMPUTE Ω | N−1 REL. | PARETO RANK |
|---|---|---|---|---|
| **K-Means Clustering** TOP | **2.852 ms** | **0.003 s** | **100%** | **Rank 1** |
| **Greedy K-Center** | 3.104 ms | 0.002 s | 100% | Rank 2 |
| **PyTorch DQN** | 2.914 ms | 21.26 s | 100% | Rank 3 |
| **Genetic Algorithm (GA)** | 3.180 ms | 0.450 s | 100% | Rank 4 |
| **Random Baseline** | 4.820 ms | 0.0001 s | 85.2% | Rank 5 |

**Statistical Significance:** Factorial ANOVA confirms significant algorithmic effect ($F(4, 55) = 89.4$, $p < 0.001$). Post-hoc Tukey HSD proves K-Means significantly outperforms GA and Random baselines.

## 7 Sustainable Decision Framework

OPERATOR GUIDE

✓ **Static WANs (N ≤ 50)**
**Deploy K-Means / Greedy K-Center:** Sub-5ms compute overhead, zero GPU requirement, minimal energy consumption, and provable latency bounds.

⚡ **Dynamic / Volatile WANs**
**Deploy PyTorch DRL / DQN:** Strictly justified when facing real-time topology churn, high-frequency link degradation, or multi-tenant traffic shifts.

## 8 Conclusions & Recommendations

KEY TAKEAWAYS

- **Decision Efficiency:** For static WANs, classical clustering delivers optimal decision quality without neural training overhead.
- **Sustainable Digital Innovation:** Eliminating unnecessary AI compute aligns directly with green networking paradigms by avoiding computational complexity taxes.
- **Practical Recommendation:** Adopt a tiered placement architecture—use deterministic heuristics as the primary operational backbone, triggering DRL adaptation only during severe network churn.



**Key References:** [1] Heller et al. (2012) *CCR* • [2] Gonzalez (1985) *TCS* • [7] Farahi et al. (2026) *IEEE TNSM*